

How can we Describe Characteristics of Waves

Types of Wave Motion

waves can be described as transverse and longitudinal.

Definition: Transverse wave has direction of vibration that is perpendicular to the direction of wave travel

#PhysicsDefinitions

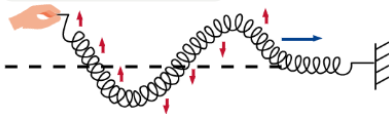


Figure 11.11 For a transverse wave, the vibration of the spring is perpendicular to the wave motion.

Definition: Longitudinal wave has a direction of vibration that is parallel to the direction of wave travel

#PhysicsDefinitions

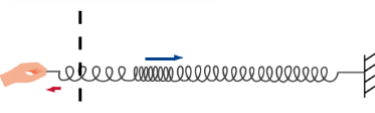
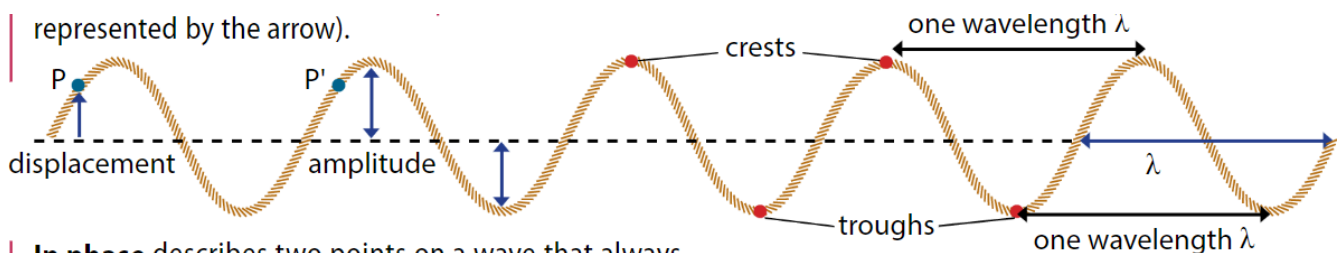


Figure 11.12 For a longitudinal wave, the vibration of the spring is parallel to the wave motion.

Examples of transverse waves: light waves, vibrating guitar string when plucked

Examples of longitudinal waves: sound waves, certain waves of an earthquake

Basic Characteristics of Waves



In phase describes two points on a wave that always

- **Displacement:** of any point on the wave is a vector pointing from its rest position to the point (Point P has displacement represented by the arrow)
- **In Phase** describes 2 points on a wave that always have same direction of motion (moving in step).
 - For example: P and P' are in phase. All crests on a wave are always in phase. All troughs on a wave are always in phase
- **Amplitude:** of a wave is its maximum magnitude of displacement from its rest position.

- **Crests**: are the highest points of a transverse wave
- **Wavelength λ** is the shortest distance between 2 successive crests or troughs
- **Troughs**: are the lowest points of a transverse wave

Dotted line $\Rightarrow$ indicates ropes resting position before wave was created

shape of the wave is called the **Waveform**

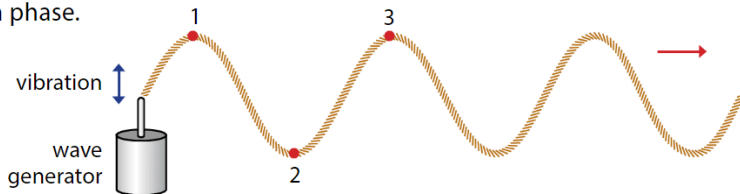
2 points along the wave are said to be in phase if they are at the same stage in a cycle.

$\Rightarrow$ implies that they always have the same direction of motion

As wave travels from left to right $\Rightarrow$ we observe that points 1 and 3 will always be in the same phase.

points 1 and 3 are said to be moving in phase:

in phase.



Points 2 and 3, while taking same time to complete each cycle $\Rightarrow$ are always in opposite directions or completely out of step.

points 2 and 3 are said to be completely out of phase.

Graphs

[Displacement-Distance Graph](#)

[Displacement-Time Graph](#)

Relationship Between Period, Wave Speed, Frequency and Wavelength

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